

Tutorial 9, Oct 1, 2025

- (1) Imp. TAs please do surface of revolution
(how to parametrize)

Please Follow the procedure on pp 140-141
of notes maintain the same
notation.

- (2) page 155 of notes: $\iint_T e^{x-y} dx dy$ problem

- (3) page 155 of notes: $\iint y^3(x^2+y^2)^{-3/2} dx dy$.

Solution partly given in notes - students have
to read before coming to tut class - doubts only
to be taken up.

- (4) Highlight the thing in blue on p167 of notes
Imp. will be used frequently.

$$0 < C < B < A$$

- (a) parametrize the curve of
intersection of $Ax^2 + By^2 + Cz^2 = \alpha$
 $A^2x^2 + B^2y^2 + C^2z^2 = \beta$.

lying in the positive octant.

Hint: Supplement it with $x^2 + y^2 + z^2 = t$

- (b) Compute $\frac{\partial(\alpha, \beta; t)}{\partial(x, y, z)}$.

- (5) Example on pp 171-172. Curve of intersection
of $x^2 + y^2 + z^2 = 4a^2$; $(x-a)^2 + y^2 = a^2$
parametrize; set up $\int_C f ds$

- (6) Discuss (students should do the calculation)
Strategy for evaluating $\iint_T y x dy dx$ page 174

(7) problem 41 p1033 TF.